

Clase Donation

```
public class Donation {  
    private String date;  
    private String document;  
    private String ngo;  
    private String projectCode;  
    private double amount;  
  
    public Donation(String date, String document, String ngo, String projectCode, double  
amount) {  
        this.date = date;  
        this.document = document;  
        this.ngo = ngo;  
        this.projectCode = projectCode;  
        this.amount = amount;  
    }  
  
    public String getDate() {  
        return date;  
    }  
  
    public String getDocument() {  
        return document;  
    }  
  
    public String getNgo() {  
        return ngo;  
    }  
  
    public String getProjectCode() {  
        return projectCode;  
    }  
  
    public double getAmount() {  
        return amount;  
    }  
}
```

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@Override

public String toString() {

    return date + ";" + document + ";" + ngo + ";" + projectCode + ";" +
String.format("%.2f", amount);

}

}

```

Clase DonationData

```

import java.util.ArrayList;
import java.util.Collections;
import java.util.Comparator;

public class DonationData {

    private ArrayList<Donation> donations;

    // a) Inicializar estructura
    public void donationDataInit() {

        donations = new ArrayList<>();

    }

    // b) Parsear string
    public Donation donationParse(String text) {

        String[] parts = text.split(";");

        String date = parts[0];

        String document = parts[1];

        String ngo = parts[2];

        String projectCode = parts[3];

        double amount = Double.parseDouble(parts[4]);

        return new Donation(date, document, ngo, projectCode, amount);

    }

    // c) Añadir donación ordenada
    public void donationDataAdd(Donation donation) {

        for (Donation d : donations) {

            if (d.getDate().equals(donation.getDate()) &&

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        d.getDocument().equals(donation.getDocument()) &&
        d.getProjectCode().equals(donation.getProjectCode())) {
            System.out.println("Donación duplicada. No añadida.");
            return;
        }
    }

    donations.add(donation);

    Collections.sort(donations, Comparator.comparing(Donation::getProjectCode));
}

// d) Obtener por posición
public String donationDataGet(int position) {
    if (position >= 0 && position < donations.size()) {
        return donations.get(position).toString();
    }

    return "Posición no válida";
}

// e) Eliminar donación
public void donationDataDel(String date, String document, String projectCode) {
    for (int i = 0; i < donations.size(); i++) {
        Donation d = donations.get(i);
        if (d.getDate().equals(date) &&
            d.getDocument().equals(document) &&
            d.getProjectCode().equals(projectCode)) {
            donations.remove(i);

            System.out.println("Donación eliminada.");
            return;
        }
    }

    System.out.println("No existe la donación.");
}

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    }

    public int size() {
        return donations.size();
    }

    public Donation getDonation(int index) {
        return donations.get(index);
    }

    public void printAll() {
        for (Donation d : donations) {
            System.out.println(d);
        }
    }

    public void removeCurrent(int index) {
        if (index >= 0 && index < donations.size()) {
            donations.remove(index);
            System.out.println("Donación eliminada.");
        }
    }
}

```

Clase Main

```

import java.util.Scanner;

public class Main {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        DonationData data = new DonationData();
        data.donationDataInit();

        // a) Añadir 4 donaciones
        data.donationDataAdd(
            data.donationParse("15/04/2023;12345678N;ACN;ACN8455;15.50"));
    }
}

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data.donationDataAdd(
    data.donationParse("16/04/2023;11111111A;ONG;ONG1234;20.00"));
data.donationDataAdd(
    data.donationParse("17/04/2023;22222222B;SOL;SOL5678;30.99"));
data.donationDataAdd(
    data.donationParse("18/04/2023;33333333C;VID;VID9999;50.00"));
// b) Añadir 2 más
data.donationDataAdd(
    data.donationParse("19/04/2023;44444444D;AAA;AAA0001;10.00"));
data.donationDataAdd(
    data.donationParse("20/04/2023;55555555E;BBB;BBB0002;60.50"));
// c) Añadir una repetida
data.donationDataAdd(
    data.donationParse("15/04/2023;12345678N;ACN;ACN8455;15.50"));
// d) Eliminar 2 existentes
data.donationDataDel("16/04/2023", "11111111A", "ONG1234");
data.donationDataDel("17/04/2023", "22222222B", "SOL5678");
// e) Eliminar una no existente
data.donationDataDel("01/01/2020", "00000000X", "AAA1111");
// MENU
int option;
int current = 0;
do {
    System.out.println("\n--- MENU ---");
    System.out.println("1. Mostrar primera donación");
    System.out.println("2. Siguiete donación");
    System.out.println("3. Donación anterior");
    System.out.println("4. Mostrar donación actual");
    System.out.println("5. Eliminar donación actual");

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System.out.println("6. Añadir nueva donación");
System.out.println("7. Imprimir todas");
System.out.println("0. Salir");
option = sc.nextInt();
sc.nextLine();
switch (option) {
    case 1:
        current = 0;
        if (data.size() > 0) {
            System.out.println(data.getDonation(current));
        }
        break;
    case 2:
        if (current < data.size() - 1) {
            current++;
        }
        System.out.println(data.getDonation(current));
        break;
    case 3:
        if (current > 0) {
            current--;
        }
        System.out.println(data.getDonation(current));
        break;
    case 4:
        if (data.size() > 0) {
            System.out.println(data.getDonation(current));
        }
        break;
```

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case 5:
    data.removeCurrent(current);
    if (current >= data.size() && current > 0) {
        current--;
    }
    break;
case 6:
    System.out.println("Introduce la donación:");
    String text = sc.nextLine();
    Donation newDonation = data.donationParse(text);
    data.donationDataAdd(newDonation);
    break;
case 7:
    data.printAll();
    break;
}
} while (option != 0);
sc.close();
}
}

```

Justificación

He utilizado un ArrayList porque:

- Es fácil añadir y eliminar elementos.
- Permite recorrer las donaciones fácilmente.
- Permite acceder por posición para el menú de navegación.